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**Oil Spill-Induced Soil Degradation and Nutrient Depletion
in Agricultural Farmlands: Evidence from Ogbodo-Isiokpo,
Niger Delta, Nigeria**

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Abstract

Oil spillage is a persistent environmental challenge in the Niger Delta, where agricultural communities depend heavily on soil resources for food production and rural livelihoods. This study assessed the effects of oil spillage on selected soil quality indicators in agricultural farmlands in Ogbodo-Isiokpo Community, Ikwerre Local Government Area, Rivers State, Nigeria. A field-based comparative approach was adopted, involving reconnaissance surveys, identification of oil-spill-affected locations, and laboratory analysis of soil samples collected from polluted and unpolluted control sites. Soil quality was assessed using pH, organic matter, nitrogen, phosphorus, and potassium as key indicators. The results revealed marked differences between contaminated and control soils. Oil-spill-affected soils recorded a mean pH of 5.20 compared with 6.50 in control soils, indicating greater acidity. Mean organic matter content was 1.74% in polluted soils compared with 3.11% in control soils. Similarly, nitrogen, phosphorus, and potassium concentrations were substantially lower in contaminated soils, with mean values of 0.08%, 4.23 mg/kg, and 0.12 cmol/kg, respectively, compared with 0.22%, 10.97 mg/kg, and 0.32 cmol/kg in control soils. The findings demonstrate that crude oil contamination is associated with deterioration of important chemical and fertility-related properties of agricultural soils. Such degradation reduces the capacity of affected soils to support sustainable crop production and increases the need for soil restoration and remediation. The study highlights the need for prompt remediation of contaminated farmlands, strengthened oil-spill prevention and monitoring, and sustained soil restoration interventions in oil-affected agricultural communities of the Niger Delta.

Keywords: oil spillage; soil degradation; agricultural land; Niger Delta; Nigeria.

Introduction

Agriculture remains a fundamental component of food security, employment and rural livelihoods, particularly in developing countries where farming provides an important source of income and subsistence for rural households. The productivity of agricultural systems is strongly dependent on the quality and functional capacity of soil. Soil properties such as pH, organic matter and the availability of essential nutrients influence nutrient cycling, plant development and the capacity of agricultural land to sustain crop production. Consequently, environmental processes that alter soil chemical and biological properties can have direct implications for agricultural sustainability. The uploaded study identifies oil pollution as one such environmental pressure, particularly in petroleum-producing regions where agricultural land occurs in close proximity to oil infrastructure.

Oil pollution is a persistent environmental problem in petroleum-producing regions. When crude oil is released into terrestrial environments, it can alter soil conditions and interfere with ecological processes that support agricultural production. The contamination of agricultural soils may affect soil structure, nutrient cycling, microbial activity and vegetation, thereby reducing the productive capacity of affected land. Evidence synthesized in the research project indicates that oil contamination can lower soil pH, suppress organic matter decomposition and reduce the availability of nitrogen, phosphorus and potassium, with consequential implications for crop production.

The Niger Delta provides an important context for examining these processes. Nigeria's petroleum industry is concentrated largely within the Niger Delta, where decades of oil exploration, production and transportation have been accompanied by recurrent environmental contamination. The region contains extensive oil and gas infrastructure, while agricultural communities continue to depend on land and other natural resources for their livelihoods. The

United Nations Environment Programme's assessment of oil-impacted areas in the Niger Delta documented extensive contamination of land and other environmental resources and emphasized the need for long-term remediation. ([UNEP - UN Environment Programme](#), 2011)

The persistence of petroleum contamination is particularly important because the effects of oil pollution may extend beyond the immediate period of a spill. UNEP's assessment of Ogoniland demonstrated that decades of oil operations had resulted in contamination of land and water resources and that some contaminated environments required extensive restoration. ([UNEP - UN Environment Programme](#)) More recent UNEP reporting also indicates that oil contamination in parts of the Niger Delta remains untreated or only partially remediated, illustrating the continuing environmental management challenge associated with petroleum pollution. ([UNEP - UN Environment Programme](#), 2006)

Within Nigeria, Rivers State represents a particularly important setting for examining the relationship between oil contamination and agricultural soil quality. The state combines extensive petroleum-related activities with substantial agricultural activity. Ogbodo-Isiokpo Community, located in Isiokpo, Ikwerre Local Government Area, is an agrarian community in Rivers State where farming constitutes an important livelihood activity. Major crops cultivated in the community include cassava, yam, maize, cocoyam, plantain and vegetables. The community's dependence on agriculture makes the quality of its soil resources particularly important for household food production and economic wellbeing.

The environmental vulnerability of Ogbodo-Isiokpo is heightened by the occurrence of oil-spill-affected locations within and around agricultural areas. The research project's spatial assessment indicates that identified spill locations were concentrated around agricultural farmlands, particularly in proximity to pipeline routes. The

study further observed an association between spill-affected locations and lower soil pH values, providing localized evidence that oil contamination is accompanied by changes in the chemical condition of agricultural soils.

Soil fertility is particularly important in this context because nitrogen, phosphorus and potassium perform distinct but complementary functions in crop production. Nitrogen supports vegetative development and chlorophyll formation, phosphorus contributes to root development and energy transfer, while potassium supports plant resistance to disease and environmental stress. The research project argues that oil contamination can interfere with soil aeration and microbial processes responsible for nutrient cycling, potentially reducing the availability of these essential nutrients to crops.

Although previous research has established a broad relationship between petroleum contamination and soil degradation, localized empirical evidence remains important because soil characteristics and contamination effects vary according to environmental and site-specific conditions. The literature reviewed in the project identifies a need for localized and comparative investigations capable of directly comparing contaminated soils with uncontaminated reference sites. In particular, Ogbodo-Isiokpo has received comparatively limited attention despite its dependence on agriculture and history of oil-spill exposure.

This study therefore focuses specifically on the effects of oil spillage on selected soil quality indicators in Ogbodo-Isiokpo Community. It compares soil conditions in oil-spill-affected and control locations using soil pH, organic matter, nitrogen, phosphorus and potassium as key indicators. By generating empirical evidence from laboratory analysis of soil samples, the study contributes localized evidence on the extent to which crude oil contamination is associated with changes in the chemical fertility status of agricultural soils. The objective is to **assess** the effects of oil spillage on selected soil quality indicators, including soil pH, organic matter, nitrogen, phosphorus and potassium.

Statement of the Problem

Oil spillage presents a significant environmental challenge for agricultural communities located within petroleum-producing regions. In the Niger Delta, the interaction between oil infrastructure and agricultural land creates particular vulnerability because contamination of soil can undermine the natural resource base upon which farming depends. The research project establishes that recurrent oil spills in the region have contributed to changes in soil chemical properties, declining soil quality and broader agricultural challenges.

The fundamental problem is that crude oil contamination can compromise the chemical and biological conditions required for productive agricultural soils. Oil-affected soils in the study project were found to have lower pH, reduced organic matter and lower concentrations of nitrogen, phosphorus and potassium than unpolluted control soils. Such changes are important because soil fertility depends on the availability and balance of nutrients necessary for plant growth. Reduced nutrient availability can consequently constrain the capacity of contaminated land to sustain agricultural production.

The problem is particularly significant in communities where agriculture constitutes a major livelihood activity. Ogbodo-Isiokpo is an agrarian community whose households cultivate crops including cassava, yam, maize, cocoyam, plantain and vegetables. Continued degradation of agricultural soils therefore has implications beyond the physical environment because deterioration of the soil resource can translate into declining agricultural productivity and increased pressure on farming households.

Despite the broader body of research on oil pollution in the Niger Delta, there remains a need for community-level evidence that demonstrates how contaminated agricultural soils differ from comparable unpolluted soils. The literature reviewed in the project identifies localized, comparative and

longitudinal research as an important gap, particularly studies that connect measurable changes in soil properties with agricultural outcomes. The absence of sufficiently localized evidence can make it difficult to determine the specific nature and magnitude of soil degradation within individual farming communities.

This evidence gap is important for environmental management and remediation. Without reliable information on the condition of contaminated soils and the specific soil properties most affected, remediation interventions may not adequately address the fertility constraints experienced by farmers. The research project consequently emphasizes the need for soil restoration, effective environmental management and remediation measures in oil-affected agricultural areas.

The problem is therefore not simply the occurrence of oil spills, but the **measurable deterioration of the soil resource following contamination and the resulting reduction in its capacity to support sustainable agriculture**. Establishing the differences between polluted and unpolluted soils is consequently essential for understanding the environmental consequences of oil spillage and providing an empirical basis for targeted soil restoration.

Accordingly, this study addresses the identified gap by comparatively assessing selected soil quality indicators pH, organic matter, nitrogen, phosphorus and potassium in oil-spill-affected and control agricultural soils in Ogbodo-Isiokpo Community. The laboratory findings provide localized empirical evidence that can inform environmental monitoring, agricultural land management and remediation planning in the community and comparable oil-producing areas of the Niger Delta.

Study Objective

The study generally aims to **assess** the effects of oil spillage on soil quality in agricultural farmlands in Ogbodo-Isiokpo Community, Rivers State, Nigeria.

Significance of the Study

The significance of this study derives from the need to understand and respond to the degradation of agricultural soils associated with oil spillage. By comparing selected soil-quality indicators between oil-spill-affected and control locations, the study provides empirical evidence that can support environmental management, agricultural planning and remediation decisions in Ogbodo-Isiokpo and other oil-affected communities.

The findings provide government agencies and environmental regulators with localized evidence of changes in soil pH, organic matter and essential nutrients associated with oil contamination. Such evidence can support more targeted environmental monitoring, assessment of contaminated agricultural land and prioritization of remediation activities. The study is particularly relevant to environmental management in the Niger Delta, where petroleum-related contamination continues to present challenges for land and natural-resource management.

For policymakers, the study provides evidence that can inform policies addressing the prevention, assessment and remediation of oil-contaminated agricultural land. Rather than treating oil spills solely as isolated pollution incidents, the findings emphasize the importance of considering their consequences for the long-term quality and productive capacity of agricultural soils.

The findings are relevant to farmers because soil fertility is fundamental to crop production. The observed differences in pH, organic matter, nitrogen, phosphorus and potassium between polluted and control soils indicate that oil contamination can compromise important soil properties required for productive agriculture. The project further reports that reduced nutrient availability has implications for

crop growth and can increase farmers' dependence on fertilizer inputs to compensate for declining soil fertility.

The study provides useful baseline information for professionals involved in contaminated-land assessment and remediation. Identifying the specific soil properties affected by oil pollution can assist in determining the type and intensity of restoration measures required. The evidence can therefore contribute to the development of remediation strategies that focus not merely on removing visible contamination but also on restoring the chemical fertility of affected agricultural soils.

The study contributes community-level empirical evidence to the growing literature on petroleum contamination and soil degradation in the Niger Delta. Its comparative assessment of polluted and control soils provides a basis for further research examining the mechanisms through which crude oil contamination alters soil properties, as well as the effectiveness of different remediation approaches. The project itself identifies the need for more localized comparative research on soil-property changes and their implications for agricultural productivity.

At the community level, the study draws attention to the environmental consequences of oil spillage for agricultural resources on which rural households depend. Ogbodo-Isiokpo is an agrarian community where farming is an important livelihood activity and where crops including cassava, yam, maize, cocoyam, plantain and vegetables are cultivated. Consequently, protecting and restoring soil quality is important not only for environmental sustainability but also for maintaining the community's agricultural production capacity and food security.

Overall, the study contributes by moving beyond general descriptions of oil pollution to provide **empirical evidence of specific changes in soil-quality indicators associated with oil-spill exposure**. This evidence strengthens the basis for targeted soil restoration, environmental monitoring and sustainable agricultural management in Ogbodo-Isiokpo and comparable oil-producing communities in the Niger Delta.

Soil Fertility Depletion Theory

The **Soil Fertility Depletion Theory** provides the most direct theoretical foundation for this article because the selected objective focuses specifically on measurable changes in soil pH, organic matter, nitrogen, phosphorus and potassium. The theory is based on the proposition that soil productivity declines when essential nutrients and fertility-supporting properties are continuously depleted or degraded without adequate replenishment. Nutrient depletion may occur through processes such as erosion, leaching, intensive cultivation and external environmental pressures, including pollution.

Applied to oil-contaminated agricultural land, the theory provides a useful explanation of how petroleum pollution can accelerate deterioration in soil fertility. Hydrocarbons introduced into soil can alter soil chemistry, interfere with microbial processes and disrupt nutrient cycling. The uploaded project specifically identifies mechanisms through which oil contamination may immobilize nitrogen, reduce phosphorus availability under acidic conditions and interfere with potassium uptake. It also identifies the reduction of organic matter as an important consequence because organic matter contributes to nutrient recycling and retention.

The theory is particularly relevant to the present study because the laboratory analysis provides direct indicators of the proposed fertility-depletion process. The project found that oil-spill-affected soils recorded lower concentrations of nitrogen, phosphorus and potassium than unpolluted control soils. Similarly, the mean organic matter content was substantially lower in polluted soils (1.74%) than in control soils (3.11%), providing empirical evidence of deterioration in a key component of soil fertility.

The theory also accommodates the role of soil pH in the fertility-depletion process. Soil pH influences nutrient solubility, microbial activity and plant nutrient uptake. The project explains that oil contamination can contribute to soil acidification, which can subsequently affect nutrient

Relevance and Limitation of the Theory

The major strength of Soil Fertility Depletion Theory for this study is that it establishes a direct connection between environmental pressure and measurable changes in soil fertility. It therefore provides a clear theoretical basis for examining differences between oil-spill-affected and unpolluted soils.

However, the theory primarily emphasizes nutrient and fertility processes and does not independently explain the wider institutional, socio-economic and ecological dimensions of oil pollution. The original project addressed these broader dimensions through Environmental Degradation Theory and the Sustainable Livelihoods Framework. Since the present article is deliberately focused on the selected soil-quality objective, Soil Fertility Depletion Theory provides the most parsimonious theoretical lens.

Concept of Oil Spillage

Oil spillage refers to the accidental, operational or deliberate release of crude oil into the environment, resulting in contamination of environmental resources and disruption of ecosystem functions. In agricultural environments, spilled crude oil can come into direct contact with soil and vegetation and consequently interfere with the physical, chemical and biological processes required for productive agricultural land. The project identifies pipeline failures, equipment malfunction, operational accidents, sabotage, illegal refining activities and transportation incidents among the potential sources of oil releases.

The environmental significance of oil spillage lies not simply in the presence of hydrocarbons but in their capacity to alter the characteristics and functioning of contaminated ecosystems. In soil environments, petroleum hydrocarbons can reduce aeration, interfere with microbial activity and alter chemical conditions. These processes can affect organic matter decomposition and nutrient cycling, thereby weakening soil fertility.

For the present study, oil spillage is conceptualized as the independent environmental **stressor**, while soil pH, organic matter and nutrient availability represent the measurable soil-quality responses.

Concept of Soil Quality

Soil quality refers to the capacity of soil to perform functions necessary for sustaining plant growth, maintaining environmental quality and supporting agricultural production. It encompasses physical, chemical and biological characteristics. The present study focuses specifically on selected chemical and fertility-related indicators: pH, organic matter, nitrogen, phosphorus and potassium.

These indicators were selected because they provide complementary information about the condition and productive potential of agricultural soil. Rather than treating soil degradation as a general or purely visual phenomenon, the study operationalizes degradation through measurable differences in these soil properties.

Soil pH

Soil pH measures the acidity or alkalinity of soil and is a major determinant of nutrient availability and microbial activity. The project identifies a pH range of approximately 6.0–7.0 as generally favourable for many crops because several essential nutrients are relatively available within this range.

Oil contamination may modify soil pH and produce more acidic conditions. Increased acidity can influence nutrient availability, microbial activity and plant development. The project further reports that oil-contaminated soils in the study area recorded lower pH values than corresponding control soils. Accordingly, lower soil pH in contaminated sites is conceptualized as an indicator of oil-spill-induced chemical degradation.

Organic Matter

Organic matter represents an important component of soil fertility because it contributes to nutrient storage and cycling, soil structure and water-holding capacity. It also provides an important substrate for soil biological activity.

Oil contamination can interfere with microbial processes responsible for decomposition and nutrient cycling. Petroleum hydrocarbons may restrict oxygen availability and suppress microbial activity, thereby disturbing the processes through which organic materials are transformed and nutrients are recycled.

The empirical evidence from Ogbodo-Isiokpo supports this conceptual relationship: polluted soils recorded a mean organic matter content of 1.74%, compared with 3.11% in control soils.

Nitrogen, Phosphorus and Potassium

Nitrogen (N), phosphorus (P) and potassium (K) are essential macronutrients required for normal crop development and agricultural productivity. Nitrogen is particularly important for vegetative growth and chlorophyll formation; phosphorus contributes to root development and energy transfer; while potassium supports plant resistance to disease and environmental stress.

Oil contamination can interfere with the availability of these nutrients by altering soil chemistry, restricting aeration and suppressing microorganisms involved in nutrient mineralization and cycling.

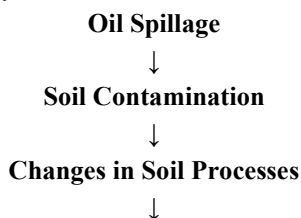
The laboratory results provide strong empirical support for this conceptual relationship. Polluted soils recorded a mean nitrogen concentration of **0.08%**, compared with **0.22%** in control soils. The project similarly reports lower phosphorus and potassium concentrations in polluted soils. These differences indicate that oil contamination is associated with a deterioration of the nutrient status of agricultural soils.

Oil Spillage and Soil Fertility

Soil fertility is the capacity of soil to supply essential nutrients in adequate quantities and proportions to sustain crop growth. Fertile agricultural soils generally require an appropriate balance of organic matter, suitable pH and adequate macronutrient availability.

The conceptual relationship proposed in this study is therefore that oil spillage acts as an environmental stressor that disrupts the chemical and biological processes supporting soil fertility. The resulting changes in pH, organic matter and NPK availability constitute measurable manifestations of soil degradation.

The conceptual model is consequently:



Soil pH | Organic Matter | Nitrogen | Phosphorus | Potassium

↓
Soil Fertility Status

This framework is consistent with the evidence from the project, which found that oil-spill-affected soils were consistently more acidic and had lower organic matter and NPK concentrations than unpolluted control soils.

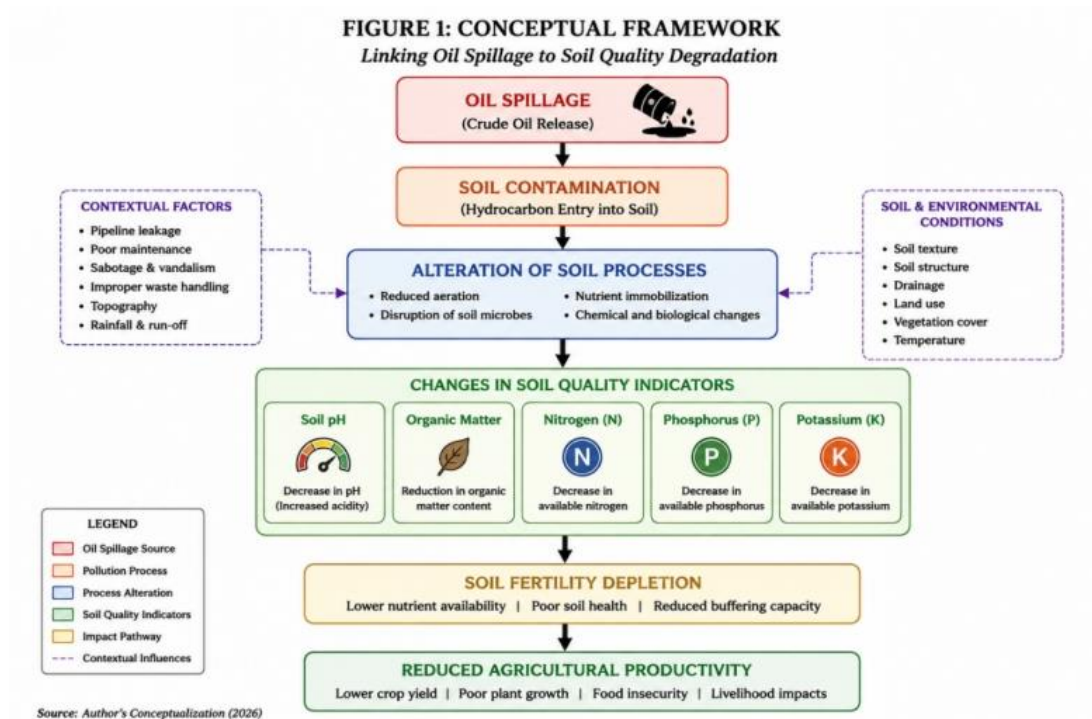


Figure 1. Conceptual Framework Linking Oil Spillage to Soil Quality Degradation

Source: Author's conceptualization based on the theoretical and conceptual relationships established in the study, 2025.

This conceptualization deliberately keeps the article focused on the selected objective. Agricultural productivity and livelihoods are treated as broader implications of soil degradation rather than additional dependent variables in the present journal article.

Study Area

The study was conducted in Ogbodo-Isiokpo Community, Isiokpo, Ikwerre Local Government Area of Rivers State, Nigeria, within the upland section of the Niger Delta. The community is located approximately between latitudes 4°93'N and 5°00'N and longitudes 6°87'E and 6°90'E, and is approximately 25 km north of Port Harcourt. Ogbodo-Isiokpo is predominantly an agrarian community, with cassava, yam, maize, cocoyam, plantain and vegetables among the major crops cultivated.

The study area was selected because of its history of oil exploration and reported oil-spill incidents, together with the importance of agriculture to local households. The occurrence of oil-affected agricultural land makes the community suitable for comparative assessment of soil quality between contaminated and non-contaminated farmlands.

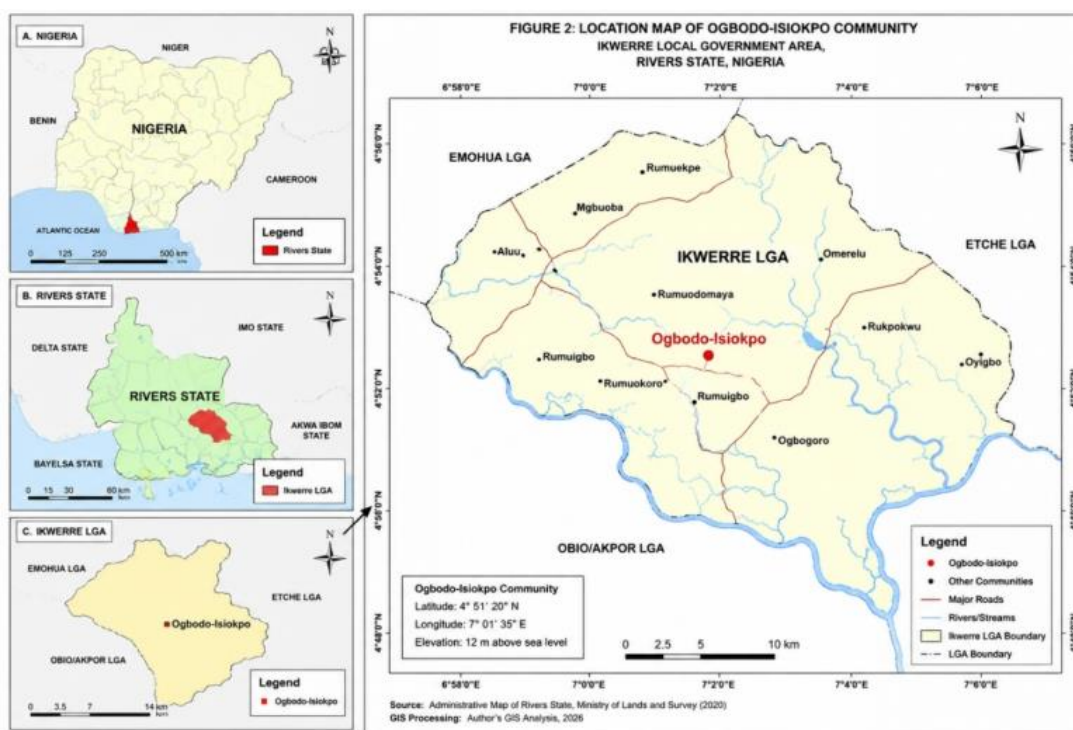


Figure 2 showing the study area

Source: GIS Lab, Geography Department Nsuk, 2025.

Research Design

The original research adopted a mixed-methods design, combining a descriptive household survey with an experimental laboratory-based component. For the present article, which focuses specifically on the selected soil-quality objective, the laboratory component constitutes the principal analytical approach, while field reconnaissance and spatial observations provide contextual information for identifying polluted and control sampling locations.

The laboratory component involved collecting and analysing soil samples from oil-polluted and non-polluted agricultural lands. The analysis focused on five soil-quality indicators: pH, organic matter, nitrogen, phosphorus and potassium.

The comparative design was particularly important because it provided a reference against which the condition of oil-spill-affected soils could be evaluated. The original methodology explicitly incorporated both polluted and non-polluted farmlands to strengthen comparison of soil-quality conditions.

Study Population

The broader study population comprised farmers and farming households in Ogbodo-Isiokpo Community. Local government records and information from community leaders estimated approximately **1,200–1,500 active farmers** in the community. Agriculture constitutes a major source of food and income, making farmers the population most directly dependent on the quality of agricultural soils.

However, because the present article is restricted to the soil-quality objective, the principal analytical population is represented by the agricultural soils sampled from oil-spill-affected and control farmlands rather than the household survey respondents.

Soil Sampling and Sampling Procedure

A total of **10 composite soil samples** were collected for laboratory analysis: five from oil-polluted agricultural farmlands and five from non-polluted control farmlands.

Soil samples were collected at two depths:

- **0–15 cm** topsoil
- **15–30 cm** subsoil

A soil auger was used for sampling. At each sampling location, three sub-samples were collected and homogenized to produce a composite sample. Composite sampling was employed to reduce variability arising from small-scale differences in soil conditions. The project indicates that the sampling procedure followed standard soil-sampling protocols recommended by the Food and Agriculture Organization. The polluted and control locations were differentiated using field reconnaissance, visible evidence of contamination, information from farmers and community stakeholders, and the spatial distribution of identified oil-spill locations. The original field investigation found that many identified spill locations were situated within or close to agricultural farmlands and particularly around pipeline routes.

Soil Sample Preparation

Following collection, the soil samples were air-dried, ground and passed through a 2-mm sieve before laboratory analysis. This preparation was undertaken to obtain relatively uniform soil material for subsequent determination of the selected physicochemical properties.

Laboratory Analysis

Five soil-quality indicators were analysed: soil pH, organic matter, nitrogen, phosphorus and potassium. These variables were selected because they represent important indicators of soil fertility and are relevant to evaluating the effects of petroleum contamination on agricultural soils.

Soil pH

Soil pH was determined using a calibrated pH meter following preparation of a 1:2.5 soil-to-water suspension.

Organic Matter

Organic matter content was determined using the Walkley–Black method, based on oxidation of organic carbon.

Nitrogen

Total nitrogen was measured using the Kjeldahl digestion method.

Phosphorus

Available phosphorus was extracted using the Bray-1 method and subsequently quantified spectrophotometrically.

Potassium

Potassium was determined using a flame photometer following ammonium acetate extraction. These analytical procedures provided quantitative measurements that enabled direct comparison of the selected soil-quality indicators between oil-spill-affected and control soils.

Data Collection

Primary environmental data were generated through field observation, soil sampling and laboratory analysis. Field observations were used to identify visible evidence of environmental degradation, including soil discoloration, vegetation stress, abandoned farmland and other indicators of oil contamination. Geographic coordinates of oil-spill locations and soil-sampling points were also obtained using a GPS device and used in the broader study's GIS analysis.

Although the broader project also employed questionnaires and key-informant interviews, these data sources are not treated as primary analytical evidence in the present article because the selected objective concerns the measured effects of oil spillage on soil quality.

Validity and Reliability of Measurements

Several procedures were adopted to enhance the reliability of the environmental measurements. Soil samples were collected systematically from both affected and non-affected agricultural areas, while composite sampling was used to reduce small-scale variability. Laboratory analyses followed standardized procedures for determining pH, organic matter, nitrogen, phosphorus and potassium.

Laboratory equipment was calibrated before analysis and standard operating procedures were followed. Selected samples were analysed in duplicate to assess the consistency and reproducibility of laboratory measurements. These quality-control procedures strengthened the reliability of the soil-quality data used in the study.

Data Analysis

The broader project used descriptive statistics, GIS analysis and laboratory measurements. Survey data were coded and analysed using **SPSS Version 25**, while spatial information was processed using GIS techniques.

For this article, analysis is restricted to the laboratory soil-quality data. The values obtained for pH, organic matter, nitrogen, phosphorus and potassium were organized according to soil condition oil-spill-affected versus unpolluted control **soils** and compared using their mean values.

The comparative analysis provides a direct assessment of the direction and magnitude of differences in the selected soil-quality indicators between contaminated and control locations.

Ethical Considerations

The broader study involved human participants through household questionnaires and therefore incorporated procedures for responsible collection of information from farming households. The present article, however, focuses on laboratory analysis of environmental samples. No individual-level personal information is reported in the soil-quality analysis.

Field activities were conducted within the study community, with sampling locations identified through field reconnaissance and community-level information. The article reports environmental measurements at aggregate sampling-site level rather than identifying individual farmers.

Methodological Summary

The methodology therefore establishes a clear analytical pathway:

Oil-spill-affected and control farmlands → systematic soil sampling → composite sample preparation → laboratory analysis → measurement of pH, organic matter, N, P and K → comparative assessment of soil quality.

This approach provides an empirical basis for determining whether measurable differences exist in important soil-quality indicators between oil-contaminated and uncontaminated agricultural soils in Ogbodo-Isiokpo.

Effects of Oil Spillage on Selected Soil Quality Indicators

The assessment of soil quality was based on a comparison of selected physicochemical properties between oil-spill-affected agricultural soils and non-polluted control soils in Ogbodo-Isiokpo. The indicators examined were soil pH, organic matter, nitrogen, phosphorus and potassium. These variables provide complementary measures of soil chemical condition and fertility status.

Table 1. Comparative Soil Quality Characteristics of Oil-Spill-Affected and Control Soils

Soil quality indicator	Oil-spill-affected soil	Control soil	Difference
Soil pH	5.20	6.50	-1.30
Organic matter (%)	1.74	3.11	-1.37
Nitrogen (%)	0.08	0.22	-0.14
Phosphorus (mg/kg)	4.23	10.97	-6.74
Potassium (cmol/kg)	0.12	0.32	-0.20

Source: Field/Laboratory Analysis, 2026

The results demonstrate a consistent pattern of poorer soil-quality conditions in the oil-spill-affected locations. All five indicators recorded lower mean values in polluted soils than in the control soils. The pattern is important because it suggests that oil contamination was not associated with deterioration of only one isolated soil characteristic; rather, the affected soils exhibited simultaneous changes in acidity, organic matter and essential nutrient availability.

Soil pH

The mean soil pH was **5.20 in oil-spill-affected soils**, compared with **6.50 in the control soils**, representing a difference of 1.30 pH units.

This difference indicates that the contaminated soils were considerably more acidic than the control soils. Soil pH is particularly important because it influences nutrient solubility, microbial activity and the availability of nutrients to plants. The project identifies a pH range of approximately 6.0–7.0 as favourable for many agricultural crops and explains that excessive acidity can constrain nutrient availability and microbial processes.

The finding therefore supports the proposition that oil contamination can modify the chemical condition of agricultural soils. The lower pH observed in the polluted locations is consistent with the conceptual framework developed for the study, in which oil contamination is expected to produce changes in soil chemistry that ultimately affect soil fertility.

Organic Matter

The mean organic matter content was 1.74% in oil-spill-affected soils, compared with 3.11% in control soils, indicating a reduction of approximately 1.37 percentage points.

Organic matter is a critical component of productive agricultural soil because it contributes to nutrient storage and cycling, soil structure and water retention. The reduction observed in the polluted soils therefore has implications for the overall fertility status of the affected farmlands.

The finding is also consistent with the proposed mechanism of oil-induced soil degradation. Petroleum contamination can interfere with soil aeration and microbial activity, thereby disrupting biological processes involved in organic matter decomposition and nutrient cycling.

The substantially higher organic matter content in the control soil suggests that the unpolluted locations retained comparatively better fertility-supporting conditions. Conversely, the lower value recorded in polluted soils provides evidence of deterioration in an important soil-quality attribute.

Nitrogen

Nitrogen showed another substantial difference between the two categories of soil. The polluted soils recorded a mean nitrogen concentration of 0.08%, whereas the control soils recorded **0.22%**.

The observed difference is important because nitrogen is essential for vegetative growth, chlorophyll formation and general plant development. A reduction in available nitrogen can therefore constrain crop growth, particularly where the soil is already subject to other fertility limitations.

The finding is consistent with the theoretical framework based on soil fertility depletion. Petroleum contamination can interfere with microbial processes responsible for nutrient transformation and cycling, potentially reducing the availability of nitrogen to plants. The lower nitrogen concentration recorded in contaminated soil consequently provides empirical support for the proposed relationship between oil pollution and deterioration of soil fertility.

Phosphorus

Phosphorus concentration was also lower in oil-spill-affected soils. The polluted soils recorded a mean value of 4.23 mg/kg, compared with 10.97 mg/kg in the control soils.

The difference of 6.74 mg/kg represents a substantial reduction in the phosphorus status of the contaminated soils. Phosphorus is important for root development and energy transfer within plants; consequently, reduced phosphorus availability can negatively influence crop establishment and development.

The observed pattern reinforces the broader finding that oil contamination was associated with a reduction in several essential soil nutrients rather than a change in only one nutrient. It also supports the conceptual pathway whereby oil contamination alters soil chemical and biological processes and subsequently reduces nutrient availability.

Potassium

Potassium concentration followed the same pattern. Oil-spill-affected soils recorded a mean potassium concentration of 0.12 cmol/kg, compared with 0.32 cmol/kg in the control soils.

Potassium performs important functions in plant regulation and contributes to plant resistance to environmental stress. The lower concentration in polluted soils therefore represents another important indication of reduced nutrient status.

Taken together with the nitrogen and phosphorus results, the potassium finding demonstrates that the effect of oil contamination extended across the three major macronutrients assessed in the study.

Overall Interpretation of the Soil Quality Results

The comparative findings reveal a consistent deterioration in all five selected soil-quality indicators in oil-spill-affected locations. The polluted soils were more acidic and contained lower concentrations of organic matter, nitrogen, phosphorus and potassium than the control soils.

This pattern provides empirical support for the study's theoretical proposition that environmental contamination can contribute to soil fertility depletion. Rather than interpreting the individual indicators independently, their combined pattern suggests a broader deterioration in the chemical fertility condition of contaminated agricultural soils.

The results are particularly significant in an agrarian community such as Ogbodo-Isiokpo, where farming households depend on agricultural land for food production and income. The project identifies cassava, yam, maize, cocoyam, plantain and vegetables among crops cultivated in the community.

The results consequently indicate that oil spillage represents more than a temporary environmental disturbance. Where contamination produces persistent changes in soil pH, organic matter and nutrient availability, the productive capacity of agricultural land may also be affected. The original project similarly concludes that oil-contaminated soils exhibited poorer fertility characteristics than control soils and that the observed changes have implications for agricultural production.

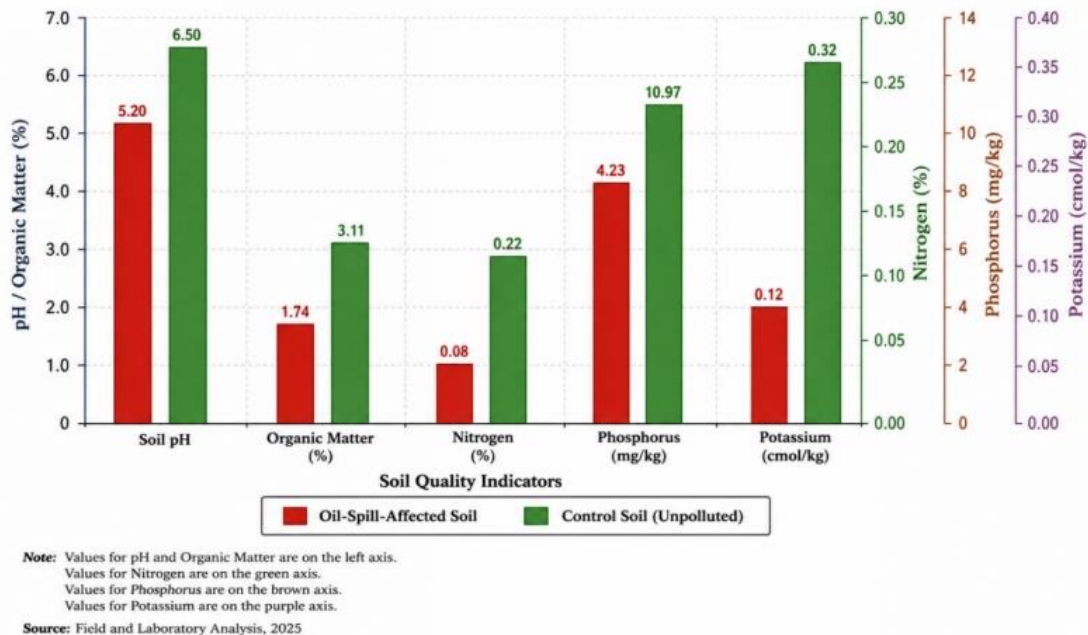


Figure 3. Comparative Mean Soil Quality Indicators in Oil-Spill-Affected and Control Soils

Source: Field/Laboratory Analysis, 2025.

The results collectively answer the selected study objective by demonstrating measurable differences in **soil pH, organic matter, nitrogen, phosphorus and potassium** between contaminated and control agricultural soils. The direction of the differences is consistent across all indicators, with the oil-spill-affected soils exhibiting poorer soil-quality characteristics.

Conclusion

This study assessed the effects of oil spillage on selected soil-quality indicators in agricultural farmlands in Ogbodo-Isiokpo Community, Rivers State, Nigeria. The assessment focused on soil pH, organic

matter, nitrogen, phosphorus and potassium, using a comparative analysis of oil-spill-affected and non-polluted control soils.

The findings demonstrate a consistent deterioration in the measured soil-quality indicators within oil-spill-affected locations. Polluted soils recorded a lower mean pH of **5.20**, compared with **6.50** in control soils. Similarly, organic matter was lower in polluted soils (1.74%) than in control soils (**3.11%**). The concentrations of nitrogen, phosphorus and potassium were also lower in contaminated soils, with values of 0.08%, 4.23 mg/kg and 0.12 cmol/kg, respectively, compared with **0.22%, 10.97 mg/kg and 0.32 cmol/kg** in the control soils.

The consistency of these differences across all five indicators suggests that oil spillage is associated with deterioration of important chemical and fertility-related properties of agricultural soils in the study area. The findings therefore provide empirical support for the study's theoretical proposition that petroleum contamination can contribute to soil fertility depletion through changes in soil chemistry, organic matter and nutrient availability.

The significance of these findings extends beyond soil chemistry because Ogbodo-Isiokpo is an agricultural community where households depend on cultivated land for food production and income. Continued deterioration of soil quality could consequently undermine the productive capacity of affected farmlands if contamination is not effectively managed.

Recommendations

Based on the findings of the study, the following recommendations are proposed:

Prioritize Remediation of Contaminated Agricultural Farmlands

Relevant government environmental agencies and responsible oil-sector operators should prioritize the identification and remediation of oil-contaminated agricultural lands in Ogbodo-Isiokpo. Remediation should address not only visible petroleum contamination but also the restoration of soil fertility characteristics affected by contamination.

Establish Regular Soil-Quality Monitoring

A systematic soil-monitoring programme should be established in oil-affected agricultural communities. Periodic testing of **pH, organic matter, nitrogen, phosphorus and potassium** would provide early evidence of deterioration and allow changes in soil condition to be tracked following remediation.

Strengthen Oil-Spill Prevention and Rapid Response

Oil companies and relevant regulatory authorities should strengthen pipeline inspection, maintenance, surveillance and rapid-response mechanisms. Preventing contamination is preferable to attempting to restore agricultural soils after prolonged exposure.

Integrate Soil Fertility Restoration into Remediation

Remediation programmes should incorporate measures specifically designed to restore soil fertility. Because the study identified reductions in organic matter and essential nutrients, restoration programmes should assess the need for appropriate soil amendments and other scientifically validated rehabilitation measures rather than focusing solely on the removal of petroleum contamination.

Improve Environmental Monitoring of Agricultural Areas

Environmental monitoring should give particular attention to agricultural lands located close to petroleum infrastructure. The study found that identified oil-spill locations occurred within or close to agricultural farmlands, making spatial monitoring important for protecting productive land resources.

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